

Student Name: _____

Class Teacher: _____



Hurlstone Agricultural High School

2021 HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 1

Mathematics Extension 1

General

Instructions

- Reading time – 3 minutes
- Working time – 40 minutes
- Write using black or blue pen.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- For Questions in Section 2, show relevant mathematical reasoning and/or calculations

Total

marks: 33

Section 1 – 3 marks

- Attempt Questions 1-3
- Allow about 5 minutes for this section

Section 2 – 30 marks

- Attempt Questions 4-6
- Allow about 35 minutes for this section

Written by 2021 HAHS Maths Faculty, rewritten by Ariq Abdullah

Section 1

Attempt Questions 1-3

Allow about 5 minutes from this section.

Question 1

The exact value of $\cos\left(\frac{\pi}{12}\right)$ is equal to the which of the following expressions?

a. $\cos\frac{\pi}{4}\cos\frac{\pi}{6} + \sin\frac{\pi}{4}\sin\frac{\pi}{6}$

c. $\cos\frac{\pi}{4}\sin\frac{\pi}{6} + \sin\frac{\pi}{4}\cos\frac{\pi}{6}$

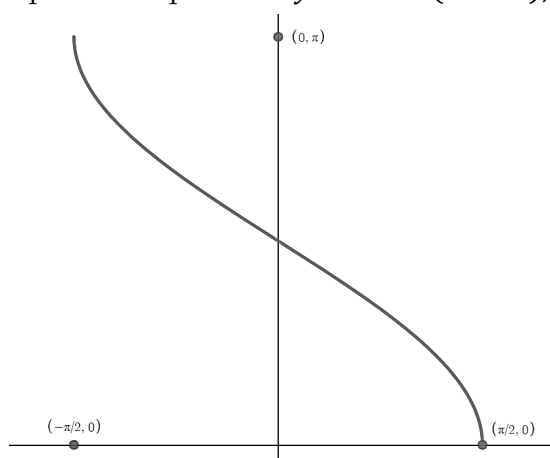
b. $\cos\frac{\pi}{4}\cos\frac{\pi}{6} - \sin\frac{\pi}{4}\sin\frac{\pi}{6}$

d. $\cos\frac{\pi}{4}\sin\frac{\pi}{6} - \sin\frac{\pi}{4}\cos\frac{\pi}{6}$

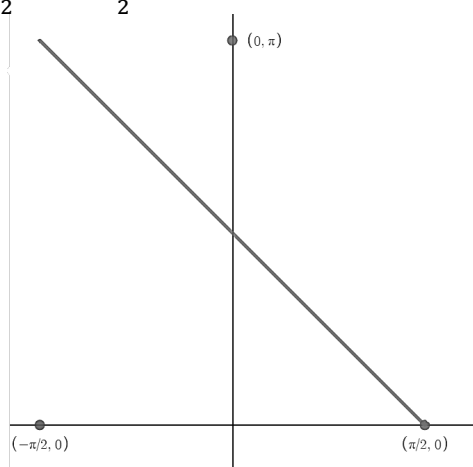
Question 2

Which graph best represents $y = \cos^{-1}(-\sin x)$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$?

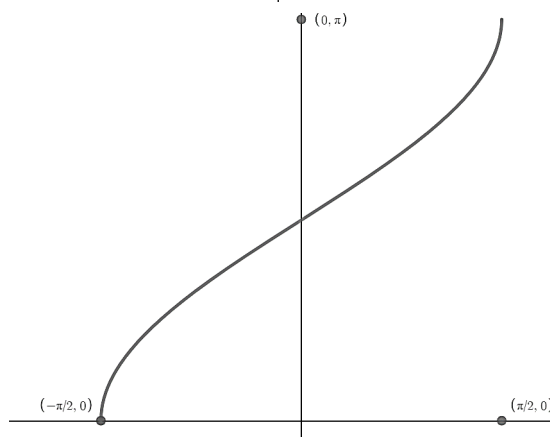
a.



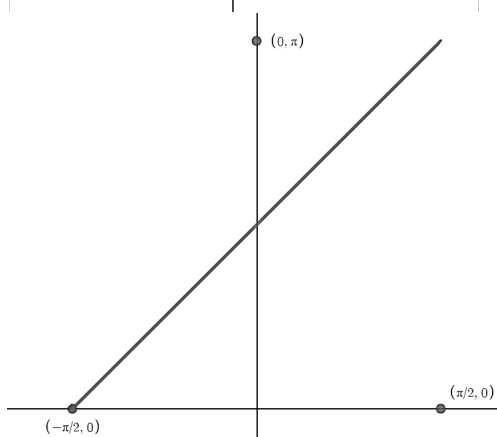
c.



b.



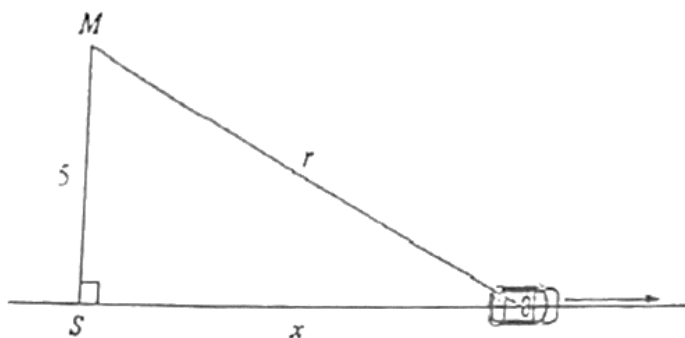
d.



Question 3

A radio transmitter M is situated 5 km from a straight road. The closest point on the road to the transmitter is S .

The distance from the car to S is x km and from the other car M is r km.



Which of the following expressions is equal to $\frac{dr}{dt}$, given that $\frac{dx}{dt} = 50$ km/h?

a. $\frac{x}{\sqrt{x^2 + 25}}$

c. $\frac{50x}{\sqrt{x^2 + 25}}$

b. $\frac{100x}{\sqrt{x^2 + 25}}$

d. $\frac{25x}{\sqrt{x^2 + 25}}$

Section 2

30 marks

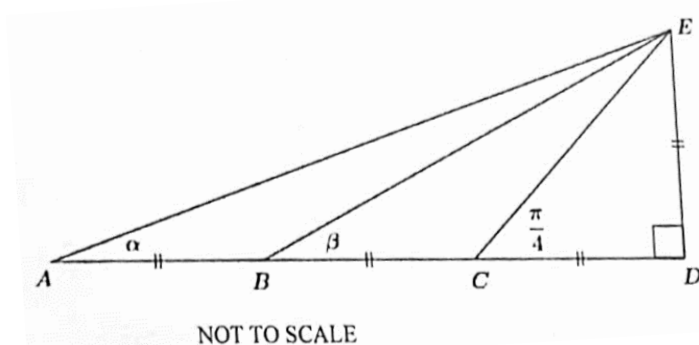
Attempt Questions 4-6

Allow about 35 minutes for this section

Question 4 (10 Marks)

Marks

- (a) Prove $\cos A - \sin 2A \sin A = \cos A \cos 2A$ 2
- (b) Rewrite the product $\sin 7x \cos 3x$ as a sum or difference of trigonometric functions. 2
- (c) Using the substitution $t = \tan \frac{\theta}{2}$, prove that $\sin \theta + \sin \theta \cot^2 \theta = \operatorname{cosec} \theta$ for $\theta \neq n\pi, n \in \mathbb{Z}$ 3
- (d) In the given figure below, $AB = BC = CD = DE$. Prove that $\alpha + \beta = \frac{\pi}{4}$. 3



Question 5 (10 marks)

Marls

- (a) A particle is moving along the x -axis. Its velocity v at time t is given by $v = \sqrt{20t - 2t^2}$ metres per second. 3
Find the acceleration of the particle when $t = 8$.
Express your answer as an exact value in its simplest form with a rational denominator.
- (b) A cup of coffee with an initial temperature 90°C is placed in a room with a constant temperature, A , of 22°C . 4
The temperature, $T^\circ\text{C}$, of the coffee after t minutes is given by $T = A + Be^{-kt}$, where A, B and k are positive constants
The temperature of the coffee drops to 60°C after 10 minutes.
How long does it take for the temperature of the coffee to drop to 30°C ?
Give your answer to the nearest minute.
- (c) A spherical balloon is being inflated at a constant rate of $200 \text{ cm}^3 \text{ s}^{-1}$. 1
- (i) Show that $\frac{dr}{dt} = \frac{1}{4\pi r^2}$, where v is the volume and r is the radius of the sphere. 1
- (ii) At what rate of the radius of the balloon increasing when the radius is 10 cm? 2

Question 6 (10 marks)

Marks

- (a) Let $f(x) = \frac{1}{2}\cos^{-1}(x)$ 2
- (i) Sketch the graph of $y = f(x)$, indicating clearly the coordinates of the endpoints of the graph
- (ii) State the range of $f(x)$. 1
- (b) Prove that $f(x) = \tan(\cos^{-1} x)$ is an odd function. 2
- (c) Write $\sin\left(2\cos^{-1}\left(\frac{1}{5}\right)\right)$ in the form $a\sqrt{b}$, where a and b are rational. 2
- (d) Find the exact values of x and y which satisfy the simultaneous equations. 3

$$\begin{aligned}\sin^{-1}x + \frac{1}{2}\cos^{-1}y &= \frac{\pi}{3} \\ 3\sin^{-1}x - \frac{1}{2}\cos^{-1}y &= \frac{2\pi}{3}\end{aligned}$$

End of Exam

Year 12	Mathematics Extension 1	Task 1	Examination 2021
Question No.4	Solutions and Marking Guidelines		
ME11-3	applies concepts and techniques of inverse trigonometric functions and simplifying expressions involving compound angles in the solution of problems		

Part	Solutions	Marking Guidelines
(a)	$\begin{aligned} \text{LHS} &= \cos A - \sin 2A \sin A \\ &= \cos A - (2 \cos A \sin A) \sin A \quad \left[\begin{array}{l} \text{Using the double-angle formulae:} \\ \sin 2A = 2 \sin A \cos A \end{array} \right] \\ &= \cos A - 2 \cos A \sin^2 A \\ &= \cos A (1 - 2 \sin^2 A) \\ &= \cos A \cos 2A \quad \left[\begin{array}{l} \text{Using the double-angle formulae:} \\ \cos 2A = 1 - 2 \sin^2 A \end{array} \right] \\ &= \text{RHS} \\ \therefore \cos A - \sin 2A \sin A &= \cos A \cos 2A \end{aligned}$	2 marks Correct solution 1 mark Substantially correct
(b)	$\begin{aligned} \sin 7x \cos 3x &= \frac{1}{2} (\sin(7x + 3x) + \sin(7x - 3x)) \\ &= \frac{1}{2} (\sin 10x + \sin 4x) \end{aligned}$	2 marks Correct solution 1 mark Substantially correct
(c)	Let $t = \tan \frac{\theta}{2}$, then $\cos \theta = \frac{1-t^2}{1+t^2}$ and $\cot \theta = \frac{1-t^2}{2t}$ Now, $\text{LHS} = \sin \theta + \sin \theta \cot^2 \theta$ $\begin{aligned} &= \frac{2t}{1+t^2} + \frac{2t}{1+t^2} \left(\frac{1-t^2}{2t} \right)^2 \\ &= \frac{2t}{1+t^2} + \frac{2t}{1+t^2} \left(\frac{(1-t^2)^2}{4t^2} \right) \\ &= \frac{2t}{1+t^2} + \frac{(1-t^2)^2}{2t(1+t^2)} \\ &= \frac{4t^2 + (1-t^2)^2}{2t(1+t^2)} \\ &= \frac{4t^2 + t^4 - 2t^2 + 1}{2t(1+t^2)} \end{aligned}$	3 marks Correct solution 2 marks Substantially correct 1 mark Correctly writes the t-formula for cotangent

$$= \frac{t^4 + 2t^2 + 1}{2t(1 + t^2)}$$

$$= \frac{(1 + t^2)^2}{2t(1 + t^2)}$$

$$= \frac{1 + t^2}{2t}$$

$$= \frac{1}{\sin \theta}$$

$$= \operatorname{cosec} \theta$$

$$= \text{RHS}$$

$$\therefore \sin \theta + \sin \theta \cot^2 \theta = \operatorname{cosec} \theta \quad \text{for } \theta \neq n\pi, n \in \mathbb{Z}$$

(d)

$$ED = DC = CB = BA \quad (\text{given})$$

$$\text{let } ED = x$$

$$\tan \frac{\pi}{4} = 1$$

$$\tan \beta = \frac{x}{2x} = \frac{1}{2}$$

$$\tan \alpha = \frac{x}{3x} = \frac{1}{3}$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}}$$

$$= 1$$

$$= \tan \frac{\pi}{4}$$

$$\alpha + \beta = \frac{\pi}{4}$$

3 marks

Correct solution

2 marks

Substantially correct

1 mark

Establishes correct expression for α and β .

2021 HSC Mathematics 1 Extension Task 1

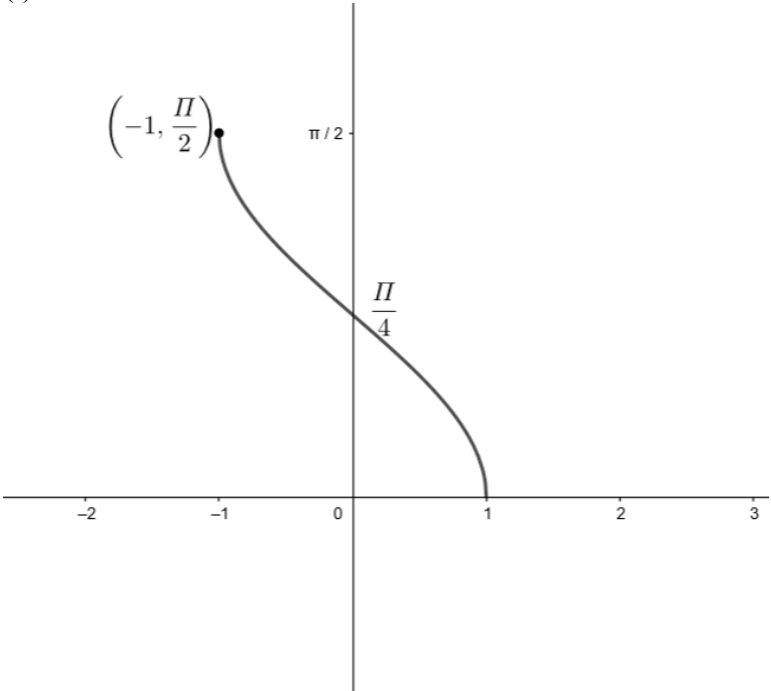
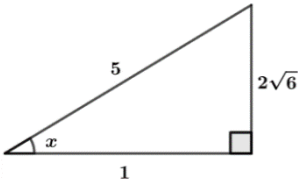
Question No. 5

Solutions and Marking Guidelines

Outcomes Addressed in this Question

ME11-4 - applies understanding of the concept of a derivative in the solution of problems, including rates of change, exponential growth and decay and related rates of change.

	Solutions	Marking Guidelines
(a)	$v = \sqrt{20t - 2t^2} = (20t - 2t^2)^{\frac{1}{2}}$ $a = \frac{dv}{dt} = \frac{1}{2}(20t - 2t^2)^{-\frac{1}{2}} \times (20 - 4t)$ $= \frac{2(5-t)}{\sqrt{20t - 2t^2}}$ $t = 8: a = \frac{2(5-8)}{\sqrt{20 \times 8 - 2(8^2)}} = \rightarrow \frac{-3\sqrt{2}}{4}$	(a) 3 marks: Correct solution, showing differentiation, substitution and simplification. 2 marks: Considerable progress. 1 mark: Some relevant progress.
(b)	$T = A + Be^{-kt} \quad A = 22; t = 0 \rightarrow T = 90$ $\therefore B = 68 \quad t = 10 \rightarrow T = 60$ $\therefore 60 = 22 + 68e^{-10k}$ $38 = 68e^{-10k}$ $e^{-10k} = \frac{19}{34}$ $\therefore k = \frac{-1}{10} \ln \frac{19}{34}$ When $T = 30$: $30 = 22 + 68e^{-kt}$ $\rightarrow \frac{2}{17} = e^{-kt}$ $t = \frac{-1}{k} \ln \frac{2}{17} \approx 37 \text{ min.}$	(b) 4 marks: Complete solution illustrating B, k, relevant equations and solution. 3 marks: Almost every factor of solution complete. 2 marks: More than 1 aspect towards solution illustrated. 1 mark: Some relevant progress. <i>Note:</i> k must be a positive value for it to fit the given equation involving $e^{(-kt)}$. If a negative sign has “gone missing” in your working out, you did not earn full marks despite giving the correct final answer.
(c)	(i) $V = \frac{4}{3}\pi r^2$ $\frac{dV}{dr} = 4\pi r^2 \quad \rightarrow \quad \frac{dr}{dV} = \frac{1}{4\pi r^2}$ (ii) $\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$ $= \frac{1}{4\pi r^2} \times 200$ $= \frac{1}{4\pi(10)^2} \times 200$ $= \frac{1}{2\pi} \text{ cms}^{-1} \quad (\approx 0.1591549)$	(c) (i) 1 mark: Correct solution justified. (ii) 2 marks: Correct answer with working. 1 mark: Correct expression and substitution into chain rule, or equivalent.

Year 12 Mathematics Extension 1		Assessment Task 1 2021
Question No. 6		Solutions and Marking Guidelines
Outcomes Addressed in this Question		
ME11-3 - applies concepts and techniques of inverse trigonometric functions and simplifying expressions involving compound angles in the solution of problems		
Solutions		Marking Guidelines
(a) (i)  (ii) For, $f(x) = \frac{1}{2} \cos^{-1}(x)$ Range is $0 \leq y \leq \frac{\pi}{2}$ (b) $ \begin{aligned} f(-x) &= \tan(\cos^{-1}(-x)) \\ &= \tan(\pi - \cos^{-1}x) \\ &= -\tan(\cos^{-1}x) \\ &= -f(x) \end{aligned} $ (c) Let $x = \cos^{-1}\left(\frac{1}{5}\right)$ $\therefore \cos x = \frac{1}{5}$ $\sin x = \frac{2\sqrt{6}}{5}$ 		Award 2 marks for the correct solution. Award 1 mark for substantial progress towards the solution Award 1 mark for the correct answer. Award 2 marks for the correct solution. Award 1 mark for substantial progress towards the solution Award 2 marks for the correct solution. Award 1 mark for substantial progress towards the solution

$$\begin{aligned}\sin\left(2\cos^{-1}\left(\frac{1}{5}\right)\right) &= \sin 2x \\ &= 2\sin x \cos x \\ &= 2 \times \frac{2\sqrt{6}}{5} \times \frac{1}{5} = \frac{4}{25}\sqrt{6}\end{aligned}$$

(d)

$$\sin^{-1} x + \frac{1}{2} \cos^{-1} y = \frac{\pi}{3} \quad \dots\dots(1)$$

$$3\sin^{-1} x - \frac{1}{2} \cos^{-1} y = \frac{2\pi}{3} \quad \dots\dots(2)$$

Add (1)+(2)

$$4\sin^{-1} x = \pi$$

$$\sin^{-1} x = \frac{\pi}{4}$$

$$\therefore x = \frac{1}{\sqrt{2}}$$

Substitute $x = \frac{1}{\sqrt{2}}$ into (1)

$$\sin^{-1} \frac{1}{\sqrt{2}} + \frac{1}{2} \cos^{-1} y = \frac{\pi}{3}$$

$$\frac{\pi}{4} + \frac{1}{2} \cos^{-1} y = \frac{\pi}{3}$$

$$\frac{1}{2} \cos^{-1} y = \frac{\pi}{3} - \frac{\pi}{4}$$

$$\frac{1}{2} \cos^{-1} y = \frac{\pi}{12}$$

$$\cos^{-1} y = \frac{\pi}{6}$$

$$\therefore y = \frac{\sqrt{3}}{2}$$

Award 3 marks for the correct solution.

Award 2 mark for substantial progress towards the correct solution.

Award 1 mark for some progress towards the correct solution.

Multiple Choice Answers

1. A

2. D

3. C